AIR-GB Lightning

Multi-path, intent-aware routing for RGB assets

Local AI proposes · deterministic code verifies · Rust routes

35% → 65%
payment success, 20 scenarios

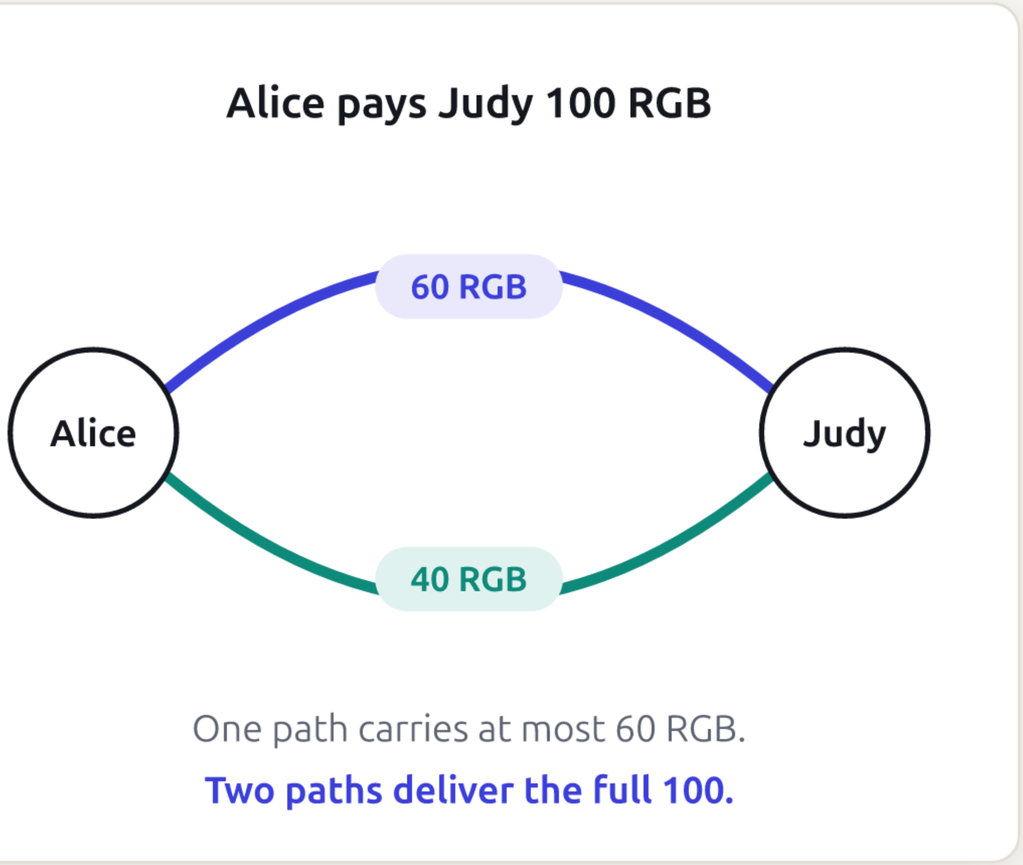
−79%
fees on payments both completed

< 1.4 ms
route planning on the node

Built on rgb-lightning-node · rust-lightning · QVAC local AI

THE OPPORTUNITY

Why multi-path matters as RGB Lightning grows



One path carries at most 60 RGB.

Two paths deliver the full 100.

1 **Combine liquidity**

Split one payment across parallel channels.

2 **Keep channels balanced**

Return flows refill channels, so later payments stay direct.

3 **Follow user intent**

Fee caps, nodes to avoid, privacy, written in plain language.

ARCHITECTURE

How a payment gets routed

Baseline stock rgb-lightning-node

7 / 20 succeeded

Payment request → One shortest path → Send

AIR-GB

13 / 20 succeeded

1 **Understand intent**
plain-language policy

→

2 **Recall & score**
local AI + payment memory

→

3 **Verify & choose**
guardrails pick a strategy

→

4 **Plan in Rust**
multi-path graph planners

LOCAL AI

Two small local models, one job each

MEMORY

EmbeddingGemma

300M parameters · 4-bit

- Saves every payment outcome
- Recalls the 4 closest matches
- Runs before each payment

past payments

top 4 matches

REASONING

Qwen3.5

0.8B parameters · 4-bit · temperature 0

- Reads the routing request
- Scores each channel
- Suggests algorithm and split

100% on the node · no cloud calls · no keys · only the node's own data

GUARDRAILS

The AI suggests. Code decides.

Model output → Payment sent

1

Locked output

The model can only name channels that really exist.

2

Grounded in words

Fees and nodes must appear in the user's own request.

3

Real balances win

Live balances override the model's channel scores.

4

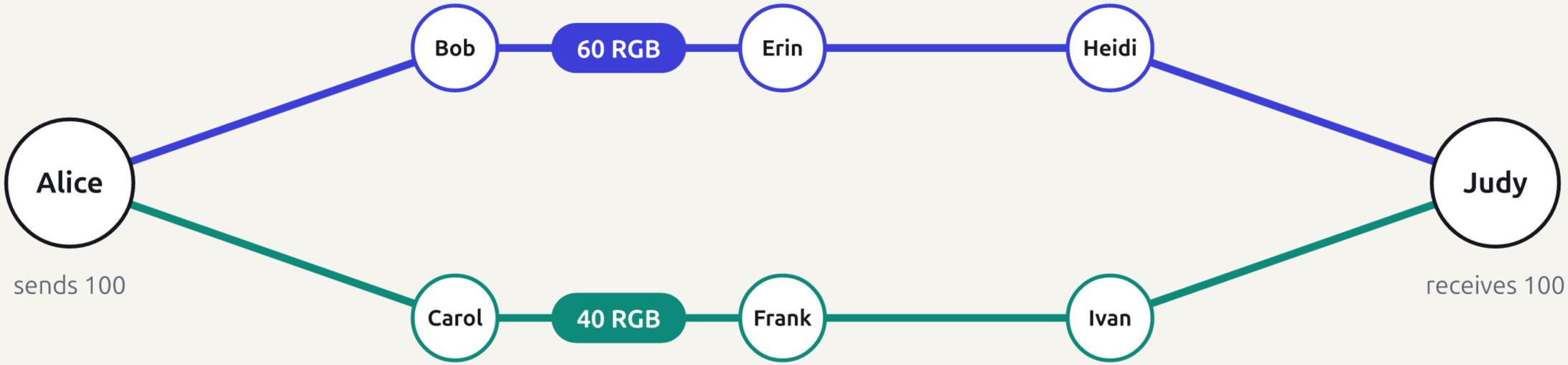
Fixed strategy rules

Split when one path can't fit; one path when privacy is asked.

A model mistake is caught before any payment is sent.

MULTI-PATH PAYMENTS

One payment, two paths, settled together



1 Check capacity

Which channels can carry the RGB?

2 Split by cost

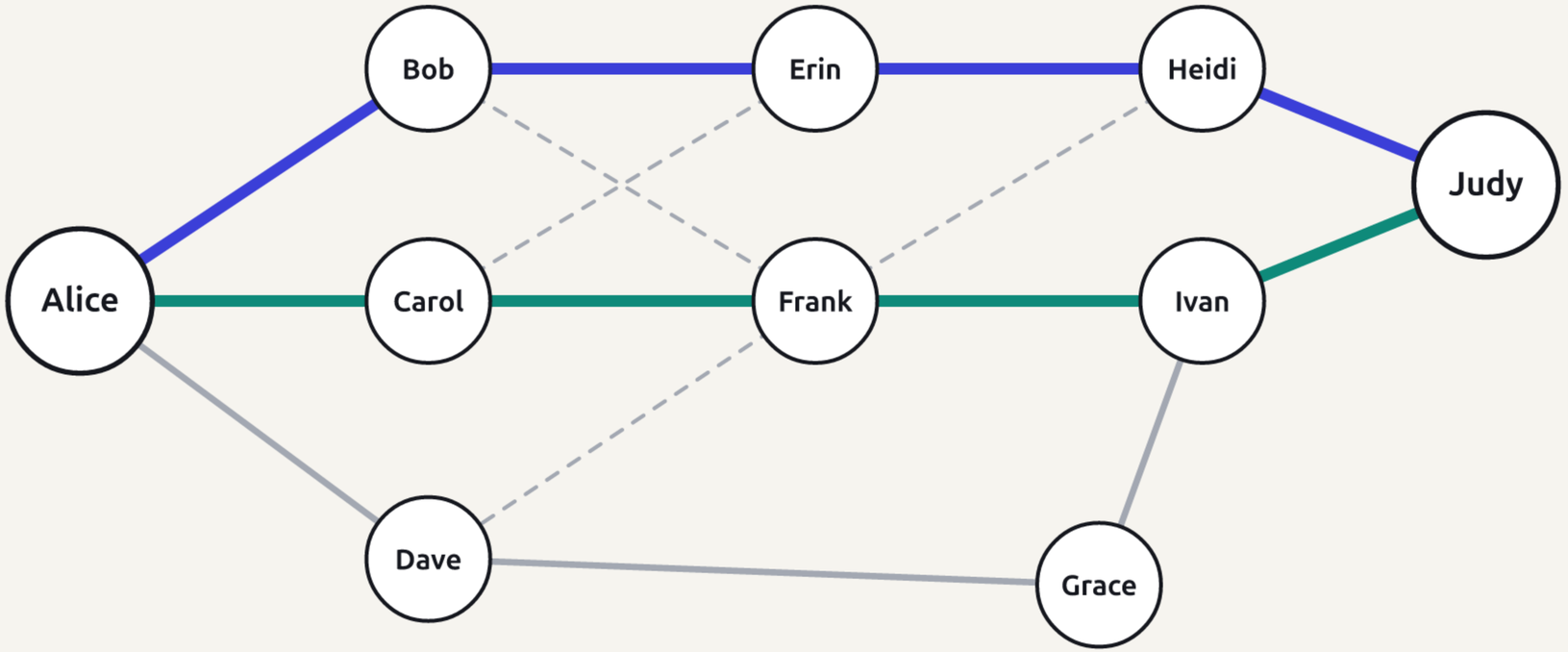
Cheaper path first, then the rest

3 Settle atomically

Judy claims only when all 100 arrive

BENCHMARK SETUP

Test network: 20 nodes, 15 RGB channels



60 RGB corridor

40 RGB corridor

other channels

high-fee cross links

+ 10 isolated nodes as negative controls

Same starting state for both runs

20 payments, run in sequence

RESULTS

Same network, same 20 payments

Payments completed +30 pts



Fees, payments both completed –79%



WHERE THE GAIN COMES FROM	BASELINE	AIR-GB
Multi-path payments (60 + 40 RGB)	0 / 3	3 / 3
RGB returned to the sender	0	190 RGB
Mid-network payments	0 / 3	3 / 3
Impossible payments rejected	4 / 4	4 / 4

LIVE DEMO

From a sentence to a multi-path RGB payment

1

Ask in plain words

“Max 1,000 msat fee, and avoid Bob.”

2

See the AI’s view

Channel scores from local payment memory

3

Send it

An RGB payment split across two paths

35% → 65% success · –79% fees · 100% local AI

Thank you. Questions welcome.